Name of Student: Rikisha Bhattacharya	**PRN: 25070122155**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 24	
TITLE: JDBC with GUI	

Problem Statement

Develop a Java GUI application integrated with JDBC to perform database operations."

Learning Objectives

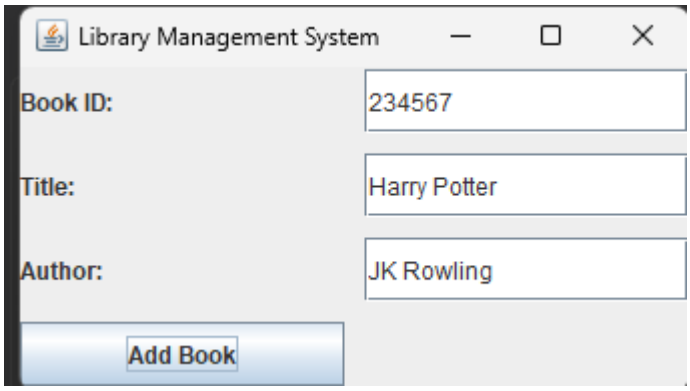
To integrate Java GUI applications with databases using JDBC.

1. **Create a Library Management System GUI connected with database**

Programme

```
39          });
40
41          setVisible(b: true);
42      }
43
44      private void addBook() {
45          String url = "jdbc:mysql://localhost:3306/rikishab";
46          String user = "root";
47          String password = "YourNewPassword123!";
48
49          String sql = "INSERT INTO books (book_id, title, author) VALUES (?, ?, ?)";
50
51          try (Connection conn = DriverManager.getConnection(url, user, password);
52               PreparedStatement pstmt = conn.prepareStatement(sql)) {
53
54              pstmt.setInt(parameterIndex: 1, Integer.parseInt(txtBookId.getText()));
55              pstmt.setString(parameterIndex: 2, txtTitle.getText());
56              pstmt.setString(parameterIndex: 3, txtAuthor.getText());
57
58              int rows = pstmt.executeUpdate();
59              if (rows > 0) {
60                  JOptionPane.showMessageDialog(this, message: "Book Added Successfully!");
61                  txtBookId.setText(t: "");
62                  txtTitle.setText(t: "");
63                  txtAuthor.setText(t: "");
64              }
65          } catch (Exception ex) {
66              JOptionPane.showMessageDialog(this, "Error: " + ex.getMessage());
67          }
68      }
69
        Run | Debug
70      public static void main(String[] args) {
71          SwingUtilities.invokeLater(() -> new LibraryGUI());
72      }
73  }
```

Output



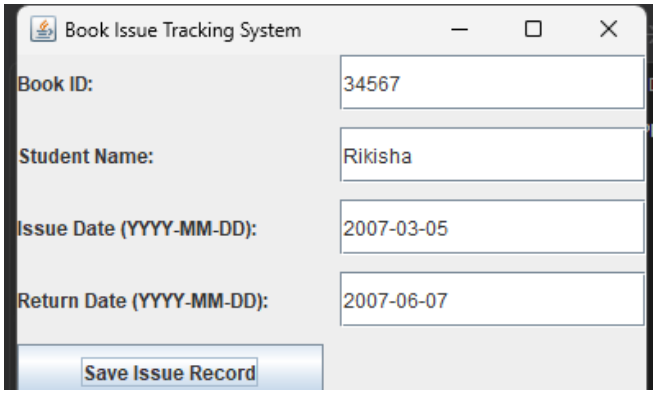
2. Create a Book Issue Tracking System GUI application using Java Swing connected to a database. The system should store book ID, student name, issue date, and return date, and allow record management.

Programme

```
Welcome   LibraryGUI.java   BookIssu   LibraryGUI
BookIssueTrackerGUI.java > BookIssueTrackerGUI
 1  import javax.swing.*;
 2  import java.awt.*;
 3  import java.awt.event.ActionEvent;
 4  import java.awt.event.ActionListener;
 5  import java.sql.Connection;
 6  import java.sql.DriverManager;
 7  import java.sql.PreparedStatement;
 8
 9  public class BookIssueTrackerGUI extends JFrame {
10      private JTextField txtBookId, txtStudentName, txtIssueDate, txtReturnDate;
11      private JButton btnSave;
12
13      public BookIssueTrackerGUI() {
14          setTitle(title: "Book Issue Tracking System");
15          setSize(width: 400, height: 250);
16          setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
17          setLayout(new GridLayout(rows: 5, cols: 2, hgap: 10, vgap: 10));
18
19          add(new JLabel(text: "Book ID:"));
20          txtBookId = new JTextField();
21          add(txtBookId);
22
23          add(new JLabel(text: "Student Name:"));
24          txtStudentName = new JTextField();
25          add(txtStudentName);
26
27          add(new JLabel(text: "Issue Date (YYYY-MM-DD):"));
28          txtIssueDate = new JTextField();
29          add(txtIssueDate);
30
31          add(new JLabel(text: "Return Date (YYYY-MM-DD):"));
32          txtReturnDate = new JTextField();
33          add(txtReturnDate);
34
35          btnSave = new JButton(text: "Save Issue Record");
36          add(btnSave);
37
38          btnSave.addActionListener(new ActionListener() {
```

```
39          @Override
40          public void actionPerformed(ActionEvent e) {
41              saveRecord();
42          }
43      });
44
45      setVisible(b: true);
46  }
47
48  private void saveRecord() {
49      String url = "jdbc:mysql://localhost:3306/library_db";
50      String user = "root";
51      String password = "password";
52
53      String sql = "INSERT INTO book_issues (book_id, student_name, issue_date, return_date) VALUES (?, ?, ?, ?)";
54
55      try (Connection conn = DriverManager.getConnection(url, user, password);
56           PreparedStatement pstmt = conn.prepareStatement(sql)) {
57
58          pstmt.setInt(parameterIndex: 1, Integer.parseInt(txtBookId.getText()));
59          pstmt.setString(parameterIndex: 2, txtStudentName.getText());
60          pstmt.setString(parameterIndex: 3, txtIssueDate.getText());
61          pstmt.setString(parameterIndex: 4, txtReturnDate.getText());
62
63          int rows = pstmt.executeUpdate();
64          if (rows > 0) {
65              JOptionPane.showMessageDialog(this, message: "Issue Record Saved Successfully!");
66              txtBookId.setText(t: "");
67              txtStudentName.setText(t: "");
68              txtIssueDate.setText(t: "");
69              txtReturnDate.setText(t: "");
70          }
71      } catch (Exception ex) {
72          JOptionPane.showMessageDialog(this, "Error: " + ex.getMessage());
73      }
74  }
```

Output



Conclusion:

This assignment successfully demonstrated the integration of Java Swing GUI applications with a MySQL database using JDBC. By building the Library Management and Book Issue Tracking systems, practical experience was gained in capturing user input, executing secure SQL queries with PreparedStatement, and providing visual feedback, ultimately solidifying the skills needed to develop interactive, data-driven desktop applications.